

CPS 200

Lecture 05

Jan 26, 2026

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Schedule

Date	Session	Quiz	Topic
Jan 5			Reading Day
Jan 7	1	No	Intro to Programming I
Jan 12			National Holiday
Jan 14	2	No	Think Python: Programming as a way of thinking, Variables and Statements
Jan 19	3	Yes	Think Python: Functions, Functions and Interfaces
Jan 21	4	Yes	Think Python: Functions and Interfaces
Jan 26	5	Yes	Think Python: Conditionals and Recursion
Jan 28	6	Yes	Think Python: Return Values
Feb 2	7	Yes	Think Python: Iteration and Search
Feb 4	8	Yes	Think Python: Strings and Regular Expressions

From Lecture 04

Loops

- Can you write a loop using the keywords `for` and `in` as well as the `range()` function? Print the numbers from 0 to 9. Discuss with partners.

Loops

- Now, try printing from 1 to 9. Discuss with partners.
- Try printing from 1 to 9 but in steps of 2 (i.e. 1, 3, 5, 7, 9). Discuss with partners.
- How would you print in steps of 3 (i.e. 1, 4, 7)? Why don't we print 9?

Loops

- Could you write a **function** named `print_to_num()` that:
 - a. Takes one **parameter** named `num`
 - b. Prints from 0 to that number (use a for loop)
- Call the function afterwards with an **argument** of your choice. Try writing it on an empty / test Colab notebook

Loops

- Let's do Bonus Question 3 from last week.
- Write a function named `print_numbers` that takes an integer argument that prints from 0 to that number. For example, `print_numbers(4)` should print the following:
 - 0
 - 1
 - 2
 - 3

Lecture 05

Booleans

- What are booleans? Can you give examples? Please discuss with your partners.

Booleans

- What would the output of each be if $x = 5$ and $y = -1$?
 - $x < y$
 - $x > y$
 - $x == y$
 - $x != y$
 - $x >= y$
 - $x <= y$

Conditionals

- Assume `x` is an integer variable with a value you do not know
 - Last week, we utilized `random.randint(-1, 1)` to assign a random variable
- Can you write some code that prints the following:
 - “`x` is less than 0” if it’s less than 0
 - “`x` is greater than 0” if it’s greater than 0
 - “`x` is equal to 0” if it’s equal to 0

Nested Conditionals

- You can **nest** conditionals inside each other
- The code on the left is functionally the same as the code on the right

```
x = random.randint(-1, 1)
if x > 0:
    print("x bigger than 0")
elif x < 0:
    print("x smaller than 0")
else:
    print("x equal to 0")
```

```
x = random.randint(-1, 1)
if x == 0:
    print("x equal to 0")
else:
    if x > 0:
        print("x bigger than 0")
    else:
        print("x smaller than 0")
```

Nested Conditionals

- Can you write a nested conditional statement that checks whether a variable is a string, then prints the length of the string only if it's over 5?
 - Hint: You would need **type()**, **str**, **len()**

Think Python

Think Python: Conditionals and Recursion

- [Chapter 05.ipynb](#)

End